

Project Proposal: TITAN

Title: Transformer-based Intelligent Task Allocation Network (TITAN) for Dual-Phase Learning and Optimal Task Scheduling

1. Introduction

Modern operating systems employ fixed heuristics and static rules for scheduling tasks, which can underperform in the face of highly dynamic, heterogeneous workloads. Recent advances in deep learning—particularly attention mechanisms—show promise for creating adaptive, context-aware schedulers. We propose TITAN, a hybrid, transformer-driven scheduler that first imitates traditional algorithms via supervised learning, then refines its policy via reinforcement learning to optimize multiple performance metrics.

2. Project Purpose and Objectives

- **Purpose:** To develop and evaluate a learning-based scheduler that delivers superior deadline adherence, reduced waiting times, and improved CPU utilization compared to classical algorithms.
- **Objectives:**
 1. Design a supervised learning pipeline to mimic Earliest Deadline First (EDF) and other rule-based schedulers.
 2. Build a transformer encoder that captures inter-task relationships and contextual features for smarter prioritization.
 3. Implement a reinforcement learning fine-tuning phase to balance multiple objectives (deadline misses, waiting time, CPU utilization).
 4. Benchmark TITAN against EDF and other baselines on synthetic and real-world workload traces.
 5. Quantify improvements in missed deadlines, average waiting time, and response time.

3. Literature Review

A survey of recent work reveals:

- Transformer architectures excel at modeling long-range dependencies in sequences (Vaswani et al., 2017).
- Prior studies apply RL to scheduling but often ignore contextual inter-task relationships or lack a cold-start imitation phase.

- Hybrid imitation-RL approaches yield faster convergence and more stable policies (Hussein et al., 2017).

4. Methodology

4.1 Data Collection and Preprocessing

- Generate diverse synthetic workloads with varying arrival rates, deadlines, and resource requirements.
- Incorporate real-world traces from cloud and embedded systems.
- Extract per-task features: arrival time, deadline, execution time, resource demand, and system state snapshots.

4.2 Model Architecture

- **Transformer Encoder:** Multi-head self-attention layers to embed each task in context of competing tasks.
- **Supervised Head:** Classifier/regressor that predicts scheduling priority scores derived from EDF or other heuristics.
- **RL Head:** Policy network that outputs scheduling actions; trained with PPO (Proximal Policy Optimization) or A2C.

4.3 Dual-Phase Training

1. **Imitation Phase:** Train on labeled data of scheduler decisions (EDF, Round Robin, etc.) with cross-entropy or MSE loss.
2. **Reinforcement Phase:** Fine-tune policy via reward that penalizes deadline misses, long waiting times, and low CPU utilization.

4.4 Evaluation Metrics**

- **Deadline Miss Rate:** Percentage of tasks finishing after their deadlines.
- **Average Waiting Time:** Mean time tasks wait in the queue before execution.
- **Response Time:** Mean time from arrival to completion.
- **CPU Utilization:** Percentage of CPU cycles actively used.

4.5 Experimental Setup

- **Baselines:** EDF, SJF, Round Robin, and recent RL-based schedulers.
- **Workload Scenarios:** Real-time streaming, cloud bursts, mixed-criticality tasks.
- **Hardware:** Simulated CPUs with configurable cores and frequencies; optionally deploy prototype on Linux kernel module for real-world tests.

5. Expected Findings

Based on preliminary experiments, TITAN is expected to:

- Reduce missed deadlines by at least 0.7% over EDF under complex conditions.
- Lower average waiting and response times by approximately 70%.
- Maintain or improve CPU utilization by learning context-dependent batching and preemption strategies.

6. Project Timeline

Phase	Duration	Milestones
Requirements & Data Gathering	1 month	Collection of workload traces and dataset ready
Model Design & Imitation	2 months	Transformer-based imitation model trained
RL Fine-Tuning	1.5 months	Policy converged on multi-objective reward
Evaluation & Benchmarking	1 month	Comparative analysis against baselines
Integration & Write-up	1.5 months	Documentation, paper draft, and prototype module

7. Budget and Resources

- **Computing:** Access to GPU servers (e.g., NVIDIA RTX 3060 Ti) and CPU clusters.
- **Software:** PyTorch, RL libraries, simulation frameworks.
- **Personnel:** 1 lead researcher, 1 PhD student, 1 research assistant.
- **Estimated Cost:** \$50,000 covering compute time, personnel stipends, and dissemination.

8. Impact and Dissemination

TITAN will demonstrate the viability of learning-based scheduling in real-time and cloud domains. Outcomes include:

- A reusable open-source scheduler library.
- A conference publication in premier systems venues (e.g., SOSP, EuroSys).
- Potential integration into OS kernels and cloud orchestrators.

9. References

- Vaswani, A., et al. (2017). "Attention is All You Need." NIPS.
- Hussein, A., et al. (2017). "Imitation and Reinforcement Learning for Policy Adaptation." IJCAI.